AnalystLab Africa – Data Science Internship Programme

Week 2 Assignment

Data Pre-processing Report

Feature Engineering & Data Pre-processing for Machine Learning
Loan Prediction Dataset

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Introduction

Following the completion of the initial data inspection and understanding phase, this project progressed to feature engineering and data pre-processing. The purpose of this phase was to transform the selected Loan Prediction Dataset into a structured format suitable for machine learning.

The dataset contains historical loan application information, including applicant characteristics, financial information, credit history, property area and the corresponding loan approval outcome. The target variable for this project is Loan_Status, representing whether a loan application was approved or rejected.

The pre-processing process focused on improving data quality, handling missing observations, creating meaningful features, encoding categorical variables, scaling numerical variables, investigating outliers and selecting features relevant to predictive modelling.

The workflow followed the requirements of the Analyst-Lab Africa Week 2 assignment, which calls for missing-value handling, duplicate detection, data-type validation, categorical encoding, feature scaling, outlier detection, feature engineering and feature selection before producing the final machine-learning-ready dataset.

Dataset Overview

The selected dataset is the Loan Prediction Dataset, provided as part of the Kaggle Loan Prediction Problem dataset.

The project focuses on the following business problem:

- Predict whether a loan application should be approved based on information available about the applicant and the loan application.

The lender is intentionally left unspecified because the dataset does not identify a particular lending institution.

Project Objective

The primary objective is:

- To develop a dataset suitable for predicting loan approval outcomes using machine-learning techniques.

Target Variable

The target variable is:

- Loan_Status

It contains two outcomes:

- Y — Loan approved
- N — Loan not approved

For machine-learning purposes, the target was converted into a numerical binary representation.

The assignment identifies the Loan Prediction Dataset as Option 2 and specifically defines its objective as preparing data for predicting whether a loan application should be approved.

Data Inspection and Initial Assessment

The dataset contains:

- 614 observations
- 13 original variables

The unit of observation is an individual loan application. The original variables include:

Variable	Description	Type
Loan_ID	Application identifier	Identifier
Gender	Applicant gender	Categorical
Married	Marital status	Categorical
Dependents	Number of dependents	Categorical
Education	Education level	Categorical
Self_Employed	Self-employment status	Categorical
ApplicantIncome	Applicant income	Numerical
CoapplicantIncome	Coapplicant income	Numerical
LoanAmount	Requested loan amount	Numerical
Loan_Amount_Term	Loan repayment term	Numerical
Credit_History	Credit history indicator	Numerical/Binary
Property_Area	Property location category	Categorical
Loan_Status	Loan approval outcome	Target

The inspection established that the dataset contained both numerical and categorical variables, requiring different pre-processing techniques.

Data Cleaning

Missing Value Assessment

Missing values were identified before any imputation was performed. The initial missing-value assessment produced the following results:

Variable	Missing Values
Loan_ID	0
Gender	13
Married	3
Dependents	15
Education	0
Self_Employed	32
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	22
Loan_Amount_Term	14
Credit_History	50
Property_Area	0
Loan_Status	0

The missing values were concentrated in categorical variables and a small number of numerical variables. Importantly, the target variable `Loan_Status` contained no missing values. Therefore, no target observations needed to be removed or imputed

Missing Value Treatment

Different imputation methods were used according to variable type and distribution.

Categorical Variables

The following variables were imputed using their **mode**:

- Gender
- Married
- Dependents
- Self_Employed
- Loan_Amount_Term
- Credit_History

Mode imputation was selected because these variables represent categories or discrete application characteristics, making the most frequently occurring category a suitable simple replacement for missing observations.

Loan Amount

- LoanAmount was imputed using the **median**.

The median was preferred because financial variables can exhibit skewed distributions and extreme values. The median is less influenced by extreme observations than the mean.

Target Variable

- Loan_Status was not imputed because it contained no missing observations.

Duplicate Record Assessment

Duplicate records were investigated using Pandas' duplicate-record check.

The result was:

- 0 duplicate records

Therefore, no observations were removed on the basis of complete row duplication. This indicates that each row represents a distinct application record within the dataset.

Data Type Validation

The original dataset contained:

- str variables representing categorical fields
- int64 variables
- float64 variables

The data types were reviewed and prepared for subsequent machine-learning transformations.

Categorical variables were retained in an appropriate categorical/string representation until the encoding stage rather than being prematurely converted into arbitrary numerical values.

Credit_History was already represented numerically and was therefore suitable for binary modelling after missing values had been addressed.

Feature Engineering

Feature engineering was conducted to create variables that could provide more meaningful information for the loan prediction problem.

Removal of Loan ID

The Loan_ID variable was removed.

Reason:

- Loan_ID is an identifier rather than a characteristic of the applicant or loan application. Its values do not represent meaningful financial, demographic or application information that should be used to predict loan approval.

Retaining an arbitrary identifier could introduce meaningless patterns into a predictive model.

Therefore:

- Loan_ID was classified as an irrelevant identifier and removed from the modelling dataset.

Column Name Standardisation

The column names were converted into a consistent lowercase snake-case format.

For example:

Loan_ID	→ loan_id
ApplicantIncome	→ applicant_income
CoapplicantIncome	→ coapplicant_income
LoanAmount	→ loan_amount
Loan_Amount_Term	→ loan_amount_term
Credit_History	→ credit_history
Property_Area	→ property_area
Loan_Status	→ loan_status

This improved consistency and made the variables easier to reference programmatically.

Creation of Total Income

A new feature called:

- total_income

was created.

It was calculated as:

- TotalIncome=ApplicantIncome+CoapplicantIncome

This feature represents the combined income associated with the loan application.

Rationale

A loan decision may reasonably involve the overall income available to support repayment rather than only the income of the primary applicant.

The engineered feature therefore provides a consolidated representation of applicant and coapplicant income.

Creation of Loan-Income Ratio

A second engineered feature was created:

- loan_income_ratio

calculated as:

$$\textbf{LoanIncomeRatio} = \frac{\textbf{TotalIncome}}{\textbf{LoanAmount}}$$

This feature represents the requested loan amount relative to the combined income of the applicants.

Rationale

While LoanAmount describes the size of the requested loan, TotalIncome describes available income.

The ratio combines these two concepts and provides an indication of the relative size of the loan request compared with the applicants' combined income.

This feature subsequently demonstrated substantial predictive importance during feature selection, ranking **second** in the Random Forest feature-importance analysis.

Categorical Feature Encoding

Machine-learning algorithms generally require categorical information to be represented numerically. The categorical variables were therefore encoded according to their structure.

Binary Encoding

Binary categorical variables were converted into numerical representations.
Variables included:

- gender
- married
- education
- self_employed
- credit_history
- loan_status

This transformation allows the binary information to be represented numerically without creating unnecessary additional columns.

Dependents was also transformed into a numerical representation, with the 3+ category represented as a higher dependent-count category.

One-Hot Encoding

property_area contained three categories:

- Rural
- Semiurban
- Urban

One-hot encoding was applied to represent the categorical property-area information numerically.

The resulting variables included:

- property_area_Semiurban
- property_area_Urban

with Rural serving as the reference category.

Rationale

One-hot encoding was appropriate because property area represents nominal categories without an inherent numerical order.

Assigning values such as:

- Rural = 0
- Semiurban = 1
- Urban = 2

could incorrectly imply an ordered relationship between the categories. One-hot encoding avoids this problem.

Numerical Feature Scaling

Numerical variables were standardized using **StandardScaler**.

The variables scaled included:

- applicant_income
- coapplicant_income
- loan_amount
- loan_amount_term
- total_income
- loan_income_ratio

Standardization transforms each variable approximately according to:

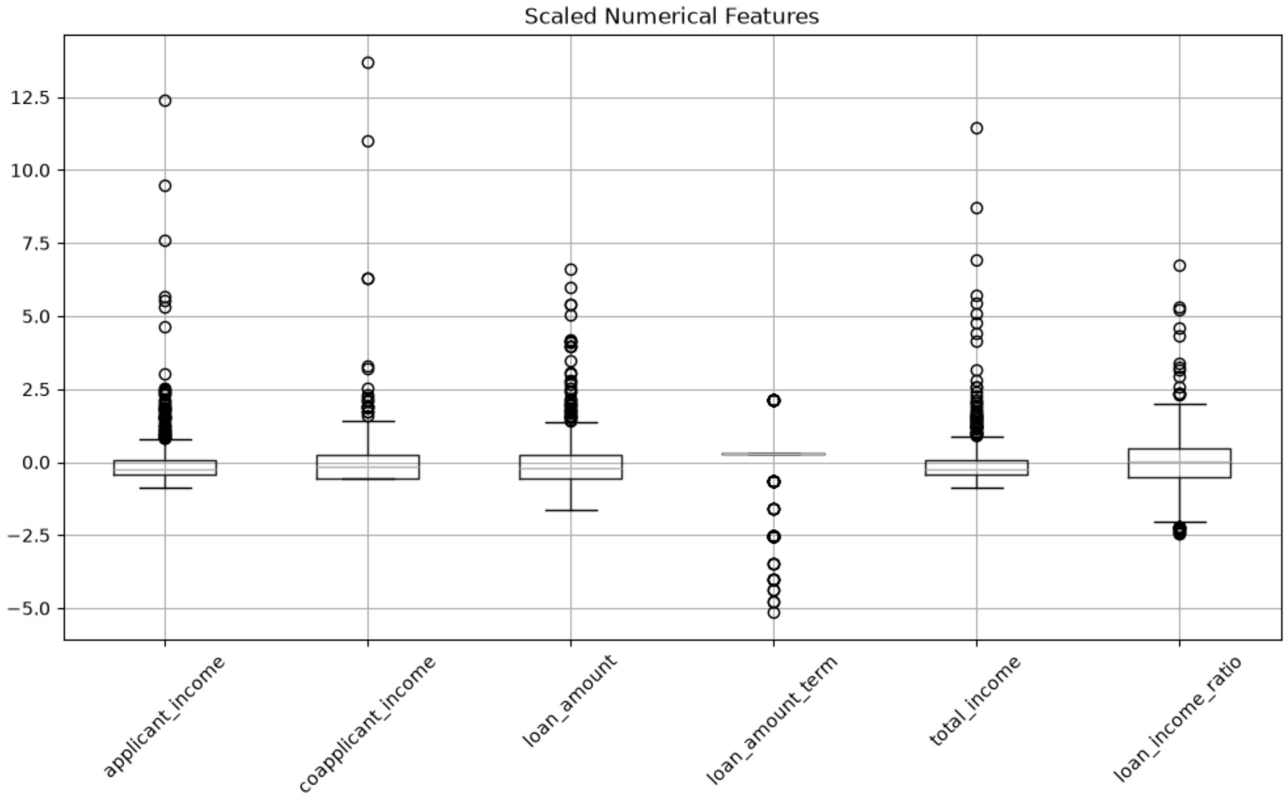
$$Z = \frac{x - \mu}{\sigma}$$

where:

- x = original value
- μ = mean
- σ = standard deviation

Following transformation, the numerical features had means approximately equal to zero and standard deviations approximately equal to one.

For example, the standardized results showed means extremely close to zero and standard deviations of approximately 1.00 across the scaled variables.



Why StandardScaler was used

The numerical variables were measured on substantially different scales.

For example, before scaling:

- Applicant income ranged from 150 to 81,000.
- Coapplicant income ranged from 0 to 41,667.
- Loan amount ranged from 9 to 700.
- Loan amount term ranged from 12 to 480.

Without scaling, variables with larger numerical magnitudes could have disproportionate influence on algorithms that are sensitive to feature scale.

StandardScaler therefore provided a common standardized scale while preserving the relative distribution of the observations.

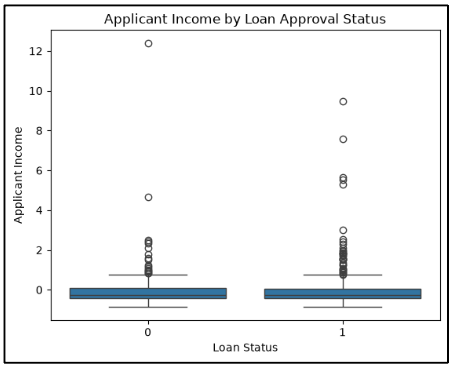
The assignment explicitly requires the use and justification of an appropriate scaling technique such as StandardScaler, MinMaxScaler or RobustScaler.

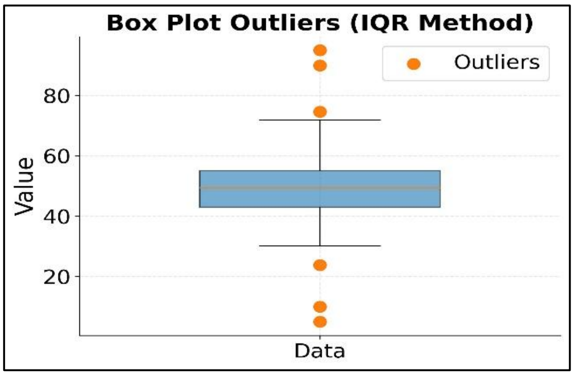
Outlier Detection

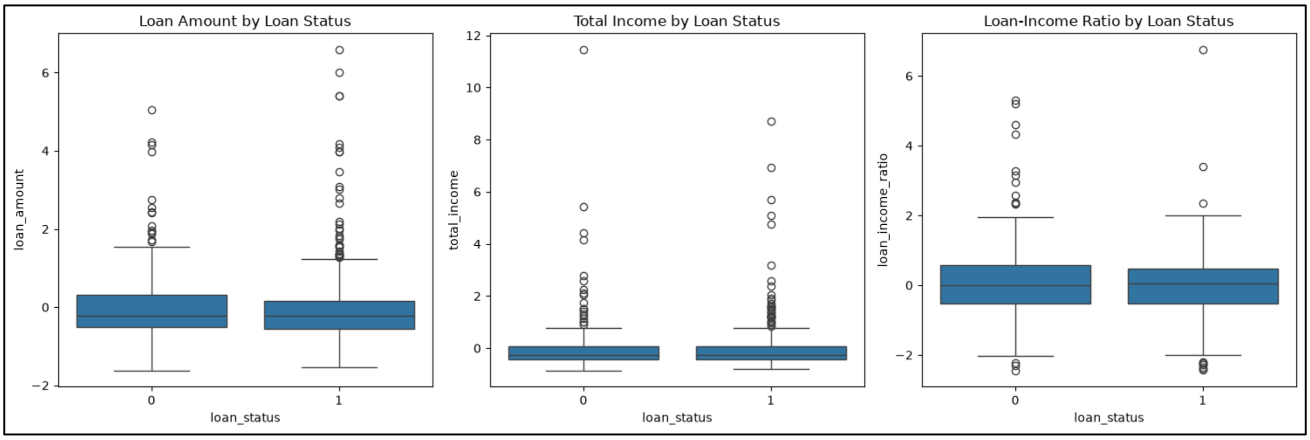
Outliers were investigated using:

- Boxplots
- Interquartile Range (IQR) analysis

The standardized numerical features were visualised using boxplots.







The analysis showed a substantial number of observations outside the conventional IQR boundaries for several variables.

The calculated results were:

Feature	IQR	Lower Bound	Upper Bound	Potential Outliers
applicant_income	0.478	-1.131	0.781	50
coapplicant_income	0.786	-1.733	1.410	18
loan_amount	0.768	-1.693	1.377	41
loan_amount_term	0.000	0.273	0.273	88
total_income	0.520	-1.223	0.857	50
loan_income_ratio	1.027	-2.076	2.03	25

Interpretation of Outliers

The presence of an observation outside the IQR boundaries does not automatically mean that the observation is erroneous.

This distinction was particularly important for the loan dataset because unusually high incomes, loan amounts or loan-to-income ratios can represent legitimate applications rather than data-entry errors.

For example, the original dataset contained:

- Applicant income as high as **81,000**
- Coapplicant income as high as **41,667**
- Loan amounts as high as **700**

These observations are unusual but cannot be classified as errors solely because they are extreme.
The assignment specifically requires outlier detection and treatment using methods such as boxplots, IQR or Z-score, while requiring the analyst to explain the chosen approach.

Decision
The identified extreme observations were **not automatically deleted**.
They were treated as potential legitimate observations and retained for subsequent modelling evaluation.
This approach avoids unnecessarily reducing the dataset or removing potentially meaningful financial cases.

Feature Selection

Feature selection was performed after preprocessing using two complementary approaches:

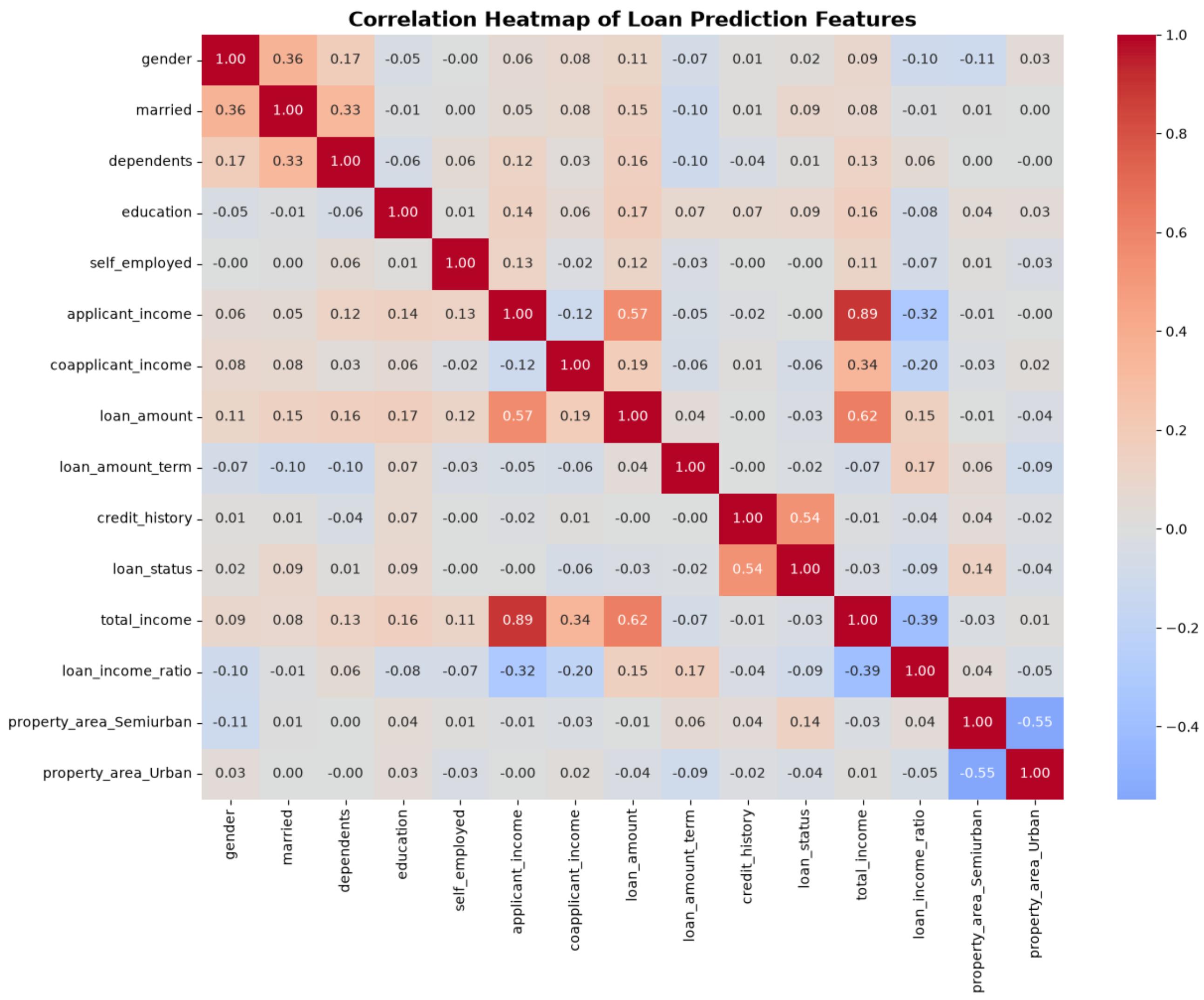
1. Correlation analysis
2. Random Forest feature importance

This directly addressed the assignment requirement to identify the most important features, highly correlated variables and features that should be removed.

Correlation Analysis

A correlation matrix and heatmap were produced to investigate relationships between the numerical variables and the target.

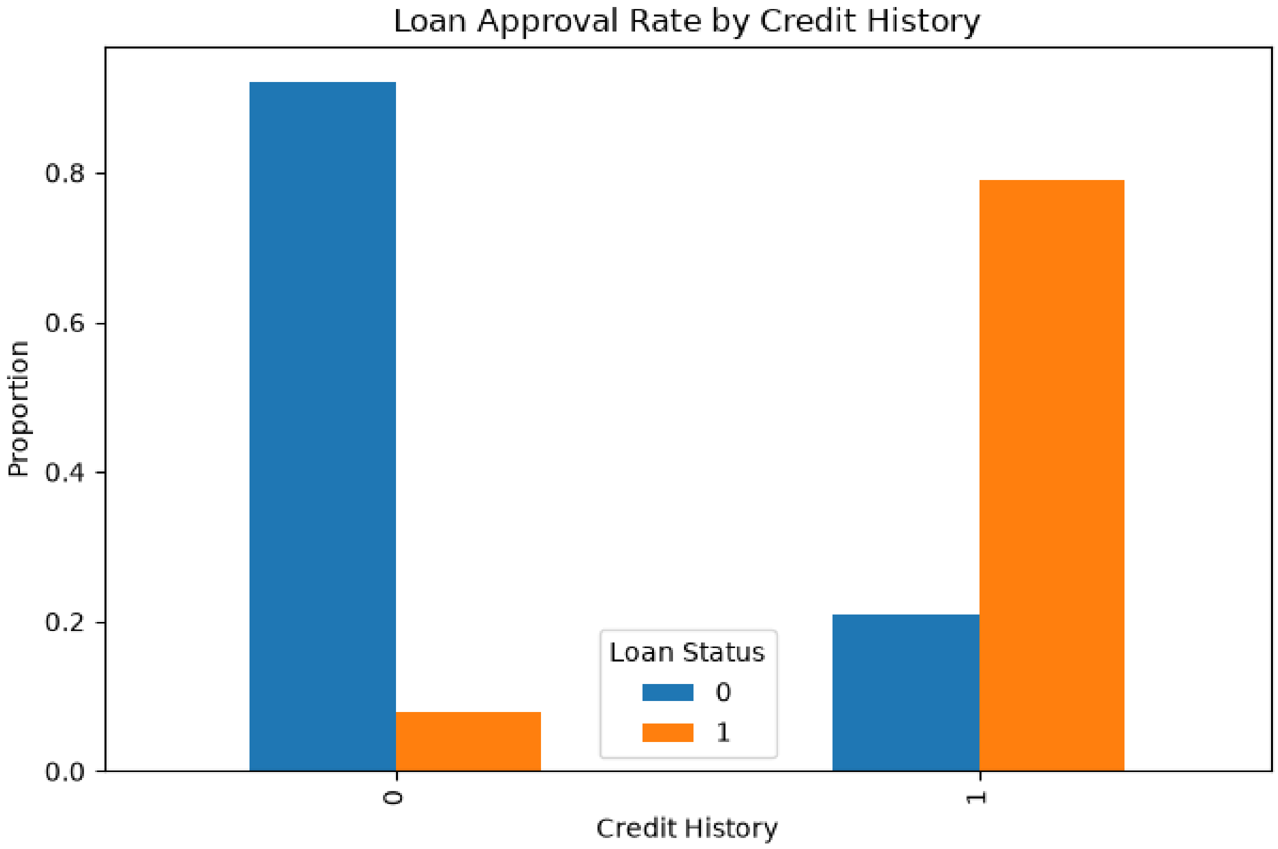
The heatmap identified the following notable relationships.



Credit History and Loan Status

r = 0.54

This was the strongest individual feature-to-target correlation observed.
This indicates a relatively strong positive association between credit history and loan approval in this dataset.



It should be interpreted as an association rather than evidence of causation.

Applicant Income and Total Income

r = 0.893

This was the strongest predictor-to-predictor correlation identified.

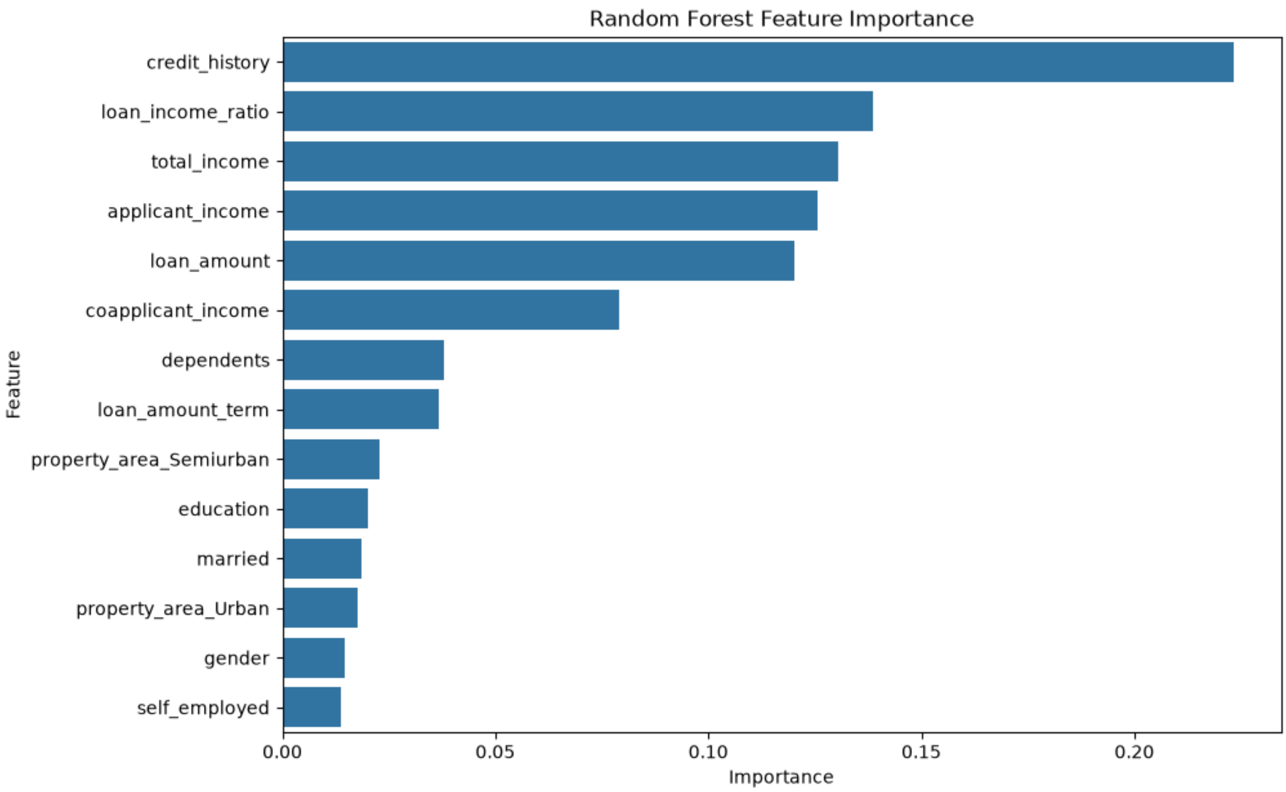
The high correlation is expected because:

TotalIncome = ApplicantIncome + CoapplicantIncome

This indicated substantial overlap between applicant_income and total_income.

Feature Importance Analysis

A Random Forest classifier was used to evaluate feature importance.



The results were:

Rank	Feature	Importance
1	credit_history	0.2232
2	loan_income_ratio	0.1386
3	total_income	0.1306
4	applicant_income	0.1257
5	loan_amount	0.1202
6	coapplicant_income	0.0791
7	dependents	0.0380
8	loan_amount_term	0.0367
9	property_area_Semiurban	0.0227
10	education	0.0202
11	married	0.0187
12	property_area_Urban	0.0177
13	gender	0.0147
14	self_employed	0.0138

Interpretation of Feature Importance

Credit History

credit_history ranked first with an importance score of **0.2232**. This is consistent with the correlation analysis, where it also demonstrated the strongest relationship with loan_status.

Therefore, it was retained as a key predictive feature.

Loan Income Ratio

loan_income_ratio ranked second with an importance of **0.1386**. Its relatively high feature importance is particularly notable because its direct correlation with loan_status was only approximately -0.09.

This demonstrates that a weak linear correlation does not necessarily mean that a feature lacks predictive value. The engineered feature was therefore retained.

Total Income

total_income ranked third with an importance of **0.1306**.

It was retained because it provides a consolidated representation of applicant and coapplicant income and is relevant to the financial context of the loan application.

Removal of Applicant Income

Although applicant_income ranked fourth in feature importance with a score of **0.1257**, it was removed from the final feature set.

The decision was based on redundancy rather than low predictive importance.

The correlation between:

- applicant_income

and:

- total_income

was:

r = 0.893

This indicates substantial overlap.

Since:

TotalIncome = ApplicantIncome + CoapplicantIncome

total_income provides a broader representation of income available to the application.

Therefore:

applicant_income was removed to reduce redundant information, while total_income was retained. This is an important distinction: Applicant Income was not removed because it was unimportant. It was removed because its information substantially overlapped with Total Income.

Remaining Feature Selection Decisions

The remaining variables were retained because their relatively low importance scores did not, by themselves, provide sufficient justification for removing them.

The final selection was:

Feature	Decision
credit_history	Retain
loan_income_ratio	Retain
total_income	Retain
applicant_income	Remove
loan_amount	Retain
coapplicant_income	Retain
dependents	Retain
loan_amount_term	Retain
property_area_Semiurban	Retain
property_area_Urban	Retain
education	Retain
married	Retain
gender	Retain
self_employed	Retain
loan_status	Target

Loan_ID had already been removed during feature engineering because it was an identifier.

Summary of Pre-processing Decisions

The complete preprocessing workflow can therefore be summarised as follows:

Stage	Action	Reason
Data cleaning	Missing values identified	Establish data quality
Data cleaning	Mode imputation for categorical/discrete variables	Preserve observations and use most frequent category
Data cleaning	Median imputation for loan_amount	Reduce sensitivity to skew/extreme values
Data cleaning	Duplicate check	Identify repeated observations
Data cleaning	0 duplicates found	No duplicate removal required
Feature engineering	Removed Loan_ID	Identifier, not predictive information
Feature engineering	Standardised column names	Improve consistency
Feature engineering	Created total_income	Represent combined income
Feature engineering	Created loan_income_ratio	Represent loan burden relative to income
Encoding	Binary variables encoded numerically	Convert binary categories into model-compatible values
Encoding	property_area one-hot encoded	Avoid imposing false ordinal relationships
Scaling	StandardScaler applied	Place numerical variables on comparable scales
Outlier detection	Boxplots + IQR	Identify unusually distant observations
Outlier treatment	Extreme observations retained	No sufficient evidence that they were errors
Feature selection	Correlation analysis	Identify relationships/redundancy
Feature selection	Random Forest importance	Identify predictive contribution
Feature selection	Removed applicant_income	Strong redundancy with total_income

Key Findings

The pre-processing analysis produced several important findings.

1. **The dataset contained manageable missingness**
   - Missing values were present in several variables, but they could be addressed through appropriate imputation without removing large portions of the dataset.

2. **No duplicate records were identified**
   - The dataset contained zero complete duplicate rows.

3. **Feature engineering added useful financial context**
   - Two new variables were created:

$$\textbf{TotalIncome = ApplicantIncome + CoapplicantIncome}$$

and:

$$\textbf{LoanIncomeRatio} = \frac{\textbf{TotalIncome}}{\textbf{LoanAmount}}$$

The second engineered feature subsequently ranked as the **second most important feature** in the Random Forest analysis.

4. **Credit History was the strongest predictor identified**

It had:

$$\mathbf{r = 0.54}$$

with the target and a Random Forest importance of:

$$\mathbf{0.2232}$$

5. Applicant Income and Total Income were highly redundant

Their correlation was:

$$\mathbf{r = 0.893}$$

leading to the removal of applicant_income.

6. Several extreme observations were present

However, extreme financial observations were not automatically considered errors and were retained.

7. Feature importance and correlation did not always agree

For example, loan_income_ratio had relatively weak linear correlation with the target but ranked second in Random Forest importance.

This demonstrates the importance of using multiple analytical approaches during feature selection.

Final Assessment of Pre-processing

The preprocessing process substantially transformed the original loan dataset from a raw collection of mixed data types into a structured dataset suitable for machine-learning preparation.

The final processed dataset has:

- Missing values addressed
- Duplicate records assessed
- Data types prepared
- Irrelevant identifiers removed
- Column names standardised
- New financial features engineered
- Categorical variables encoded
- Numerical variables standardised
- Potential outliers investigated
- Redundant features assessed
- Feature importance evaluated
- A preliminary final feature set established

The assignment's ultimate requirement is to produce a final dataset ready for machine learning and clearly explain every preprocessing step applied.

At the conclusion of Part 4, the dataset is therefore **ready to proceed to the final machine-learning dataset preparation stage**, subject to a final validation of its dimensions, data types, missing values and target structure.

Conclusion

This data preprocessing exercise demonstrated the transition from raw loan application data to a structured dataset suitable for predictive modelling.

The process involved more than simply cleaning missing values. It required evaluating the meaning of variables, engineering financially relevant features, selecting appropriate encoding and scaling methods, investigating unusual observations and determining whether individual variables provided unique predictive information.

The analysis identified credit_history as the most influential feature, while loan_income_ratio emerged as an important engineered feature. A strong correlation between applicant_income and total_income provided evidence for removing the former to reduce redundancy.

The resulting dataset provides a substantially cleaner and more structured foundation for the next stage of the project: constructing and validating the final machine-learning dataset.